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Spk2VidNet: A Hierarchical Recurrent Architecture for High-Fidelity Video Reconstruction from Long Spike-Camera Streams

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Abstract

Spike camera is a neuromorphic vision sensor with ultra-high temporal resolution, capable of capturing fast-moving scenes by firing a stream of binary spikes. However, its relatively low spatial resolution limits the acquisition of fine-grained visual details, motivating research on spike camera super resolution (SCSR). Existing SCSR methods typically operate on fixed-length spike sequences, where the accessible information is restricted to a local temporal neighborhood. Moreover, spike fluctuations hinder reliable intensity extraction. Both factors affect the performance of SCSR. To address these issues, we propose a hierarchical recurrent network named Spk2VidNet to reconstruct high-fidelity high resolution image sequences from low resolution spike data. To mitigate fluctuations, Spk2VidNet progressively exploits temporal correlations within spike stream to enhance feature representation by hierarchically enlarging temporal receptive fields. Within the recurrent block, we introduce an alignment module that leverages the motion consistency among multiple frames to jointly estimate and mutually refine inter-frame motions, achieving more accurate temporal alignment. In addition, we develop a fusion module to adaptively integrate neighboring aligned features based on multi-scale similarity for robust feature aggregation. We further propose a segment-wise training with state transfer strategy to efficiently model long-term dependencies with limited GPU memory. Experiments on synthetic and real-captured spike data demonstrate that Spk2VidNet achieves state-of-the-art performance.

1. Introduction

High-speed imaging has attracted growing interest with the emergence of applications such as robotics [7, 18] and autonomous driving [6, 8]. Conventional frame-based camera records light intensity by accumulating incoming photons over a fixed exposure time, but it often suffers from motion blur when capturing dynamic scenes.

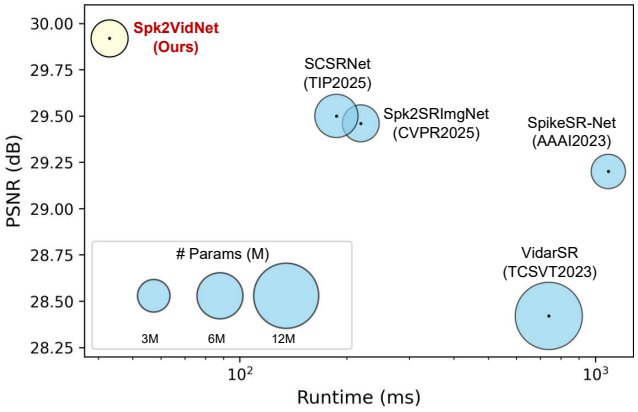


Figure 1. Model comparison of PSNR ($\times 4$), runtime, and model size. Our model achieves favorable results in terms of accuracy and efficiency.

In recent years, spike camera has emerged as an innovative vision sensor. It mimics the working mechanism of biological retina and records intensity information with ultra-high temporal resolution (*e.g.*, 40,000Hz), demonstrating great potential for high-speed vision tasks. Specifically, each pixel of spike camera independently and continuously accumulates photons, and generates spikes whenever a predefined threshold is reached. Unlike event camera [1, 9, 11, 15, 16, 20, 21] that capture *relative* changes of light intensity, the "accumulation-and-fire" principle of spike camera enables it to record *absolute* light intensities over time with fine granularity. By firing a stream of binary spikes, spike cameras capture comprehensive intensity information of the visual scene.

One limitation of spike camera lies in its relatively low

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spatial resolution, which affects its applicability in scenarios that demand fine-grained visual details. To address this issue, spike camera super resolution (SCSR) [30, 31, 33, 38, 41] has emerged as an active research topic, aiming to reconstruct high resolution (HR) images from low resolution (LR) binary spike data. Benefiting from its extremely high temporal resolution, spike camera generate a large number of spike frames within a short period (*e.g.*, 40,000 frames per second). However, existing SCSR methods typically operate on a short, fixed-length segment (*e.g.*, 101 frames [30]) of spike sequence, where only limited temporal information within a local neighborhood can be accessed. Besides, due to the randomness of photon arrivals, quantization effect from spike readout, and thermal noise from electronic circuits, spike signals suffer from fluctuations that hinder reliable intensity extraction. These issues collectively affect the performance of SCSR.

To address these challenges, we propose a hierarchical recurrent network termed Spk2VidNet to reconstruct high resolution image sequences from long spike streams. To mitigate fluctuations, Spk2VidNet gradually exploits temporal correlations within spike streams by hierarchically enlarging temporal receptive fields, enabling progressive refinement of feature representations. Specifically, considering short-term temporal correlations, we first extract an initial intensity representation for each time point from its temporally neighboring spike signals. Then, we perform propagation by aligning and aggregating nearby temporal features to improve representation. Building upon this, we further enlarge the temporal receptive field via temporal dilation and conduct a second-layer propagation, enabling the network to capture richer and more reliable temporal dependencies and thereby further enhance feature representation.

Within the recurrent propagation block, we introduce a multi-frame consistency guided alignment (MFCA) module, which leverages the motion consistency captured by spike stream with ultra-high temporal resolution. MFCA jointly estimates and mutually refines the motion among multiple frames, facilitating more accurate motion estimation and temporal alignment. Besides, we propose a content-aware modulation fusion (CMF) module that adaptively aggregates neighboring aligned features based on multi-scale similarity, enabling effective integration of correlated information while suppressing misaligned signals.

Leveraging its recurrent architecture, Spk2VidNet sequentially propagates information along the temporal dimension to model long-term dependencies, exploiting rich intensity cues in long spike sequences for improved SR reconstruction. However, training directly on long spike sequences is infeasible due to limited GPU memory. To address this, we propose a segment-wise training with state transfer strategy. Specifically, the long spike sequence is divided into multiple shorter segments, and Spk2VidNet is trained on each segment sequentially. For segments from the same spike sequence, the last several states from the preceding segment are detached and transferred to the subsequent segment, enabling Spk2VidNet to model long-term dependencies while maintaining controlled memory consumption. This strategy enables Spk2VidNet to handle arbitrarily long spike sequences. Experiments on both synthetic and real-captured spike data show that the proposed model achieves state-of-the-art performance and delivers high-quality super resolution results.

2. Related Work

Video Super Resolution. Video super resolution aims to reconstruct high resolution images from low resolution input by leveraging temporal complementary information across frames [2–4, 12, 13, 17, 23, 26–28]. A major line of research focuses on aligning adjacent frames with the reference frame to reconstruct a single HR output, where the alignment module plays a crucial role. Earlier methods [2, 17, 26] explicitly estimate optical flow to perform frame alignment. More recent developments have shifted towards implicit alignment in feature domain, where techniques such as dynamic filtering [13], deformable convolutions [27, 28], and attention mechanisms [12, 23] have been explored for motion compensation. Another category of approaches applies recurrent neural networks to exploit long-term temporal dependencies across multiple frames. FRVSR [22] uses the previous SR frame to recover the subsequent frame. RBPN [10] generates the frames through an iterative refinement process. BasicVSR [3] introduces bidirectional recurrent states, which is later extended by BasicVSR++ [4] with second-order grid propagation for enhanced performance.

Spike Camera Image Reconstruction. Spike camera image reconstruction aims to reconstruct intensity images from binary spike streams [5, 29, 35, 36, 39, 40, 42, 43, 45]. Earlier approaches primarily leverage the temporal statistical properties of spike stream for reconstruction. Zhu et al. [45] recover the intensity simply by counting the number of spikes within a temporal window or deducing from the interval between adjacent spikes. Zhao et al. [40] propose a hierarchical filtering framework that leverages both short- and long-range temporal correlations. Deep learning approaches have made notable progress in this domain. Spk2ImgNet [39] marked the first convolutional architecture specifically tailored for spike inputs and demonstrates impressive performance. Subsequently, WGSE [36] incorporates wavelet-based representation into CNN pipelines. BSF [43] designs a multi-granularity alignment method to enhance reconstruction quality. More recently, STIR [5] proposes a spatio-temporal interactive reconstruction network with low complexity.

Spike Camera Super Resolution. Spike camera super res-

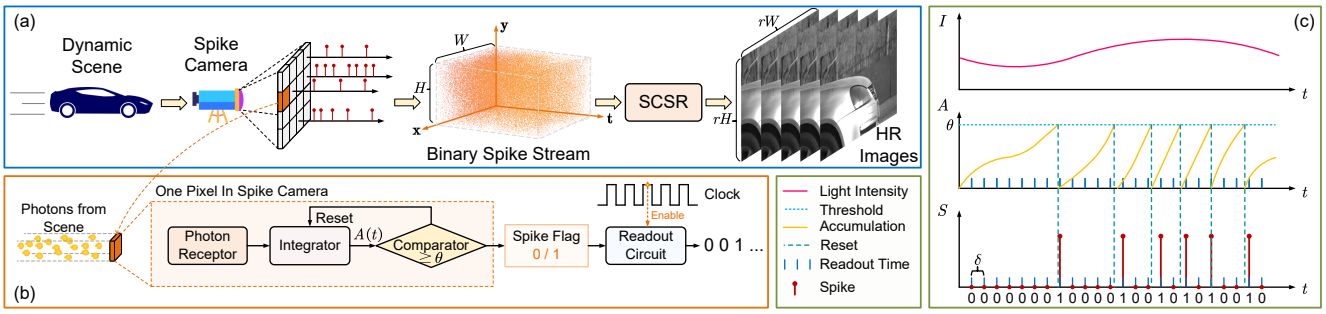


Figure 2. (a) Spike camera super resolution (SCSR). (b) Working mechanism of spike camera. (c) The relationship between light intensity $I(\mathbf{x}, t)$, the voltage $A(\mathbf{x}, t)$ accumulated by the integrator, and the spike sequence $\{S(\mathbf{x}, u)\}_{u=1}^{N}$ output by pixel $\mathbf{x}$.

olution (SCSR) focuses on generating high resolution images from low resolution binary spike frames. MGSR [38] models the relationship between light intensities and each spike cycle based on motion information and formulates an optimization objective, which is solved via gradient descent to generate HR images. VidarSR [33] proposes an end-to-end network that exploits multi-level feature learning mechanism for SCSR. Zhao et al.[41] introduce SpikeSR-Net, an optimization-inspired network with iterative architecture. Wang et al. [31] present Spk2SRImgNet, which leverages temporal similarity to mitigate disturbances in the transform domain for robust spike representation. SCSRNet [30] designs a multi-frame sequential alignment module to enhance temporal alignment accuracy and recover finer spatial details. While these methods have demonstrated notable progress, they typically operate on fixed-length spike sequences, where the accessible temporal information is restricted to a local neighborhood, thereby limiting SCSR performance.

3. Spike Camera

Working Mechanism. Spike camera contains a pixel array of size $H \times W$. Each pixel independently and continuously accumulates photons from the scene and outputs spikes to indicate the arrival of a certain amount of photons. As shown in Fig. 2(a), (b), each pixel consists of three components. The photoreceptor continuously receives photons and converts them into electric charges through photoelectric conversion. The integrator accumulates the electric charges, while the comparator checks whether the voltage exceeds a predefined threshold θ. When the accumulated voltage exceeds θ, the pixel sets up a spike flag and releases the charges in the integrator, starting a new accumulation cycle. The voltage accumulated by the integrator at pixel $\mathbf{x}$ at an arbitrary time t can be formulated as [40]:

$$A(\mathbf{x}, t) = \int_0^t \eta I(\mathbf{x}, u) du \mod \theta, \tag{1}$$

where $I(\mathbf{x}, u)$ denotes the arrival rate of photons received by pixel $\mathbf{x}$ at time u, η denotes photoelectric conversion rate. The readout circuit, driven by a clock signal with period δ, periodically checks the spike flag and outputs binary data at timestamps $\delta, 2\delta, 3\delta, ...$ When detecting a spike flag, the circuit outputs "1", otherwise, it outputs "0". The output of spike camera is a 3D binary spike matrix $\{S(u)\}_{u=1}^{N}$ with size $H \times W \times N$, where N denotes the temporal length. $S(u) \in \{0, 1\}^{H \times W}$ denotes the spike frame at time u, and $S(\mathbf{x}, u) \in \{0, 1\}$ denotes the spike value at coordinate $\mathbf{x}$ and time u. The above accumulation-and-fire cycle is illustrated in Fig. 2(c).

Basic Light Intensity Inference. Based on the working principle of spike camera, a single spike does not convey the light intensity. Rather, it indicates the arrival of a certain amount of photons. The stronger the light intensity, the shorter the time needed to accumulate to a fixed threshold; conversely, weaker light intensity corresponds to a longer accumulation cycle. TFI (Texture from Interval) [45] infers instantaneous light intensity based on this property. TFI first derives the interval between two temporally adjacent spikes that cover the spatiotemporal point $(\mathbf{x}, u)$ by:

$$\begin{aligned}\Lambda(\mathbf{x}, u) = \min\{k \mid S(\mathbf{x}, k) = 1, k \geq u\} - \\ \max\{k \mid S(\mathbf{x}, k) = 1, k < u\}.\end{aligned} \tag{2}$$

Then, TFI infers the light intensity of pixel $\mathbf{x}$ at time u by:

$$I(\mathbf{x}, u) = \frac{\gamma}{\Lambda(\mathbf{x}, u)}, \tag{3}$$

where γ denotes the maximum dynamic range of reconstruction.

4. Method

4.1. Overall Framework of Spk2VidNet

In this paper, we develop an end-to-end trainable hierarchical recurrent spike camera super resolution network named Spk2VidNet, as illustrated in Fig. 3, to reconstruct high resolution image sequences $\{\mathcal{I}_i\}_{i=0}^{N-1}$ from a long spike stream $\{S(u)\}_{u=t_0-w}^{t_{N-1}+w} \in \mathbb{B}^{H \times W \times L}$. Here, $\mathcal{I}_i \in \mathbb{R}^{rH \times rW \times 1}$ represents the reconstructed HR image at time t_i, $\mathbb{B}$ denotes

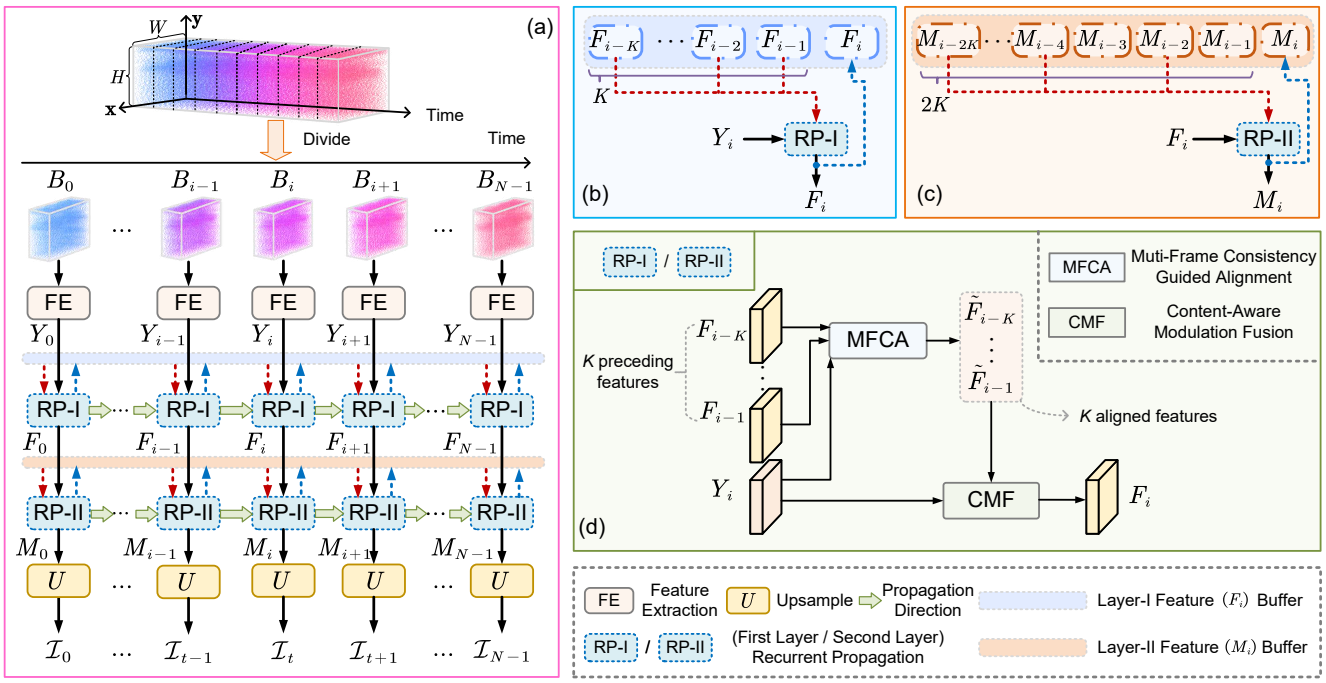


Figure 3. (a) Overall architecture of hierarchical recurrent spike camera super resolution network Spk2VidNet. (b) First-layer recurrent propagation. It propagates K preceding features to current time point. (c) Second-layer recurrent propagation. It propagates K preceding features to current time point, where the features are sampled with a temporal dilation factor of 2. (d) Illustration of recurrent propagation, using first-layer propagation RP-I as an example. The operation for RP-II is analogous, with $\{M_{i-2n}\}_{n=1}^{K}$ and F_i as input.

the binary domain, L is the temporal length of input spike stream, and r denotes SR scale.

Since one binary spike frame does not directly convey light intensity, intensity information must be inferred from contextual cues. We therefore divide the long input spike stream into N short-term spike blocks $\{B_i\}_{i=0}^{N-1}$ along the temporal dimension in an overlapping sliding-window manner, where $B_i = \{S(u)\}_{u=t_i-w}^{t_i+w} \in \mathbb{B}^{H\times W\times(2w+1)}$ denotes the spike block centered at time t_i. Exploiting short-term temporal correlations, we extract basic intensity features $\{Y_i\}_{i=0}^{N-1}$ from each block using several convolutional layers, which serve as the initial representations.

Due to randomness of photon arrivals, quantization effect from spike readout, and thermal noise from electronic circuit, spike signals exhibit fluctuations. To handle this, we perform a two-layer propagation with progressively enlarged temporal receptive field (Fig. 3(b),(c)) to generate refined features $\{F_i\}_{i=0}^{N-1}$ and $\{M_i\}_{i=0}^{N-1}$, respectively. In each recurrent propagation block (RP-I / RP-II), a multi-frame consistency guided alignment (MFCA) module and a content-aware modulation fusion (CMF) module are employed to effectively aggregate correlated temporal information, as shown in Fig. 3(d). Finally, $\{M_i\}_{i=0}^{N-1}$ is upsampled via pixel shuffle operation [24] to reconstruct HR images $\{\mathcal{I}_i\}_{i=0}^{N-1}$.

4.2. Hierarchical Propagation with Enlarged Temporal Field

To leverage rich information in long spike sequences and mitigate fluctuations, we design a hierarchical recurrent architecture that performs two-layer propagation with enlarged temporal receptive fields (as shown in Fig. 3(a–c)), enabling gradual feature refinement and enhancement.

In first layer propagation (refer to Fig. 3(a)(b)), the model takes the extracted basic representations $\{Y_0, Y_1, \cdots, Y_{N-1}\}$ as input and sequentially updates them to produce improved features $\{F_0, F_1, \cdots, F_{N-1}\}$. At each moment t_i, RP-I leverages the temporal correlation between current input Y_i and K preceding features $\{F_{i-1}, F_{i-2}, \ldots, F_{i-K}\}$ to extract temporally consistent signals, suppress disturbances, and generate F_i.

In second layer propagation (refer to Fig. 3(a)(c)), the model propagates $\{F_i\}_{i=0}^{N-1}$ to generate a sequence of further enhanced representations $\{M_i\}_{i=0}^{N-1}$. Specifically, at each moment t_i, RP-II takes current feature F_i along with K preceding features $\{M_{i-2}, M_{i-4}, ..., M_{i-2K}\}$ sampled with a temporal dilation factor of 2 as input to produce M_i, thereby expanding the temporal receptive field and capturing richer long-range temporal dependencies.

The two propagation modules, RP-I and RP-II, share the same architectural design (see Fig. 3(d)) but have independent parameters and different inputs. Here, we take RP-I as an example. RP-I takes current representation Y_i

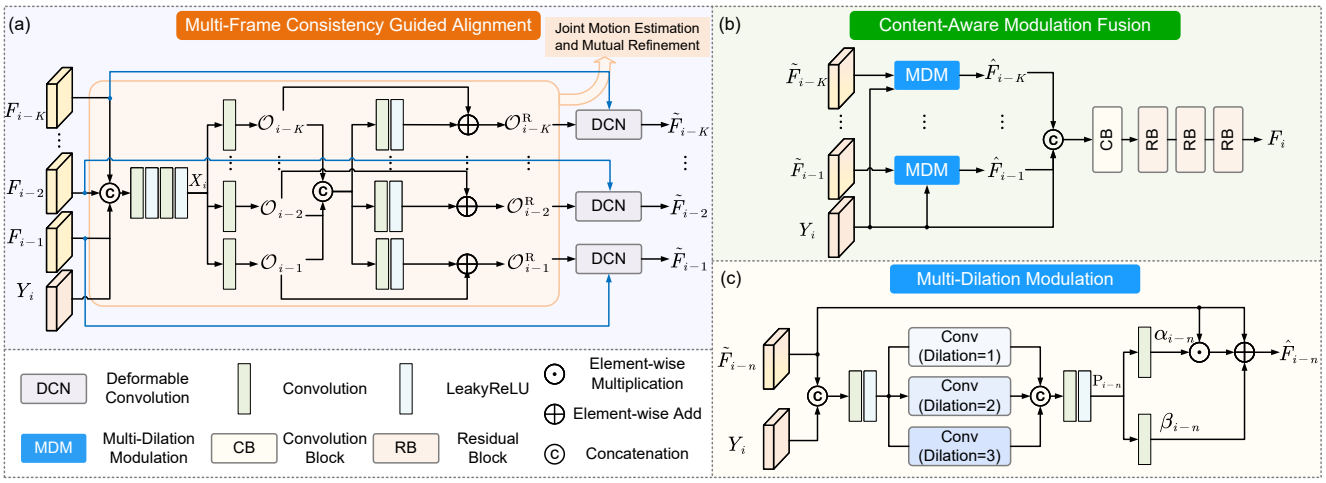


Figure 4. (a) Multi-frame consistency guided alignment (MFCA) module. (b) Content-aware modulation fusion (CMF) module. (c) Multi-dilation modulation (MDM) submodule.

and K preceding features $\{F_{i-n}\}_{n=1}^{K}$ as input. First, these features are fed into the multi-frame consistency guided alignment (MFCA) module, which aligns the neighboring features $\{F_{i-n}\}_{n=1}^{K}$ with Y_i to produce aligned features $\{\tilde{F}_{i-n}\}_{n=1}^{K}$. $\{\tilde{F}_{i-n}\}_{n=1}^{K}$, together with Y_i, are then fed into the content-aware modulation fusion (CMF) module, which adaptively fuses temporally correlated information to generate improved feature F_i. In the following, we describe the operation within RP-I in detail; RP-II follows the same procedure and is omitted for brevity.

4.3. Multi-Frame Consistency Guided Alignment

To exploit highly related but unaligned features for feature enhancement, we align the K preceding features with the current one. Benefiting from high temporal resolution of spike camera, spike stream captures dynamic motion process with fine temporal granularity. Thus, consecutive features exhibit strong motion coherence and consistency. To exploit this correlation, we design a multi-frame consistency guided alignment (MFCA) module (Fig. 4(a)), which jointly estimates and mutually refines the motion fields of the K preceding features with respect to the current feature, enabling accurate temporal alignment.

Given current input Y_i and features $\{F_{i-n}\}_{n=1}^{K}$ propagated from preceding moments, MFCA first jointly estimates the motion information among them. Specifically, MFCA concatenates $\{F_{i-K}, \ldots, F_{i-1}, Y_i\}$ along the channel dimension and extracts motion cues:

$$X_i = m([F_{i-K}, \ldots, F_{i-1}, Y_i]), \tag{4}$$

where $[\cdot,\cdot]$ denotes the concatenation operation and $m(\cdot)$ denotes several convolutional layers. Then, the K preliminary motions are respectively derived from X_i:

$$\mathcal{O}_{i-n} = f_n(X_i), \quad n = 1, \ldots, K, \tag{5}$$

where $\mathcal{O}_{i-n}$ represents the motion between F_{i-n} and Y_i, $f_n(\cdot)$ denotes convolution operation. Subsequently, leveraging the temporal consistency of these motions, MFCA updates preliminary motion estimates by incorporating mutual motion refinement signals ($n \in \{1, \ldots, K\}$):

$$\mathcal{O}^{\mathrm{R}}_{i-n} = \mathcal{O}_{i-n} + h_n([\mathcal{O}_{i-1}, \mathcal{O}_{i-2}, ..., \mathcal{O}_{i-K})]), \tag{6}$$

where $h_n(\cdot)$ denotes convolutional layers for motion interaction. The obtained K refined motion information $\{\mathcal{O}^{\mathrm{R}}_{i-n}\}_{n=1}^{K}$ is then used to align the corresponding neighboring features with the current feature:

$$\tilde{F}_{i-n} = \mathrm{DCN}(F_{i-n}, \mathcal{O}^{\mathrm{R}}_{i-n}), \quad n = 1, \ldots, K, \tag{7}$$

where $\mathrm{DCN}(\cdot)$ denotes deformable convolution [27, 28]. The joint motion estimation and mutual refinement effectively leverage the motion correlation within spike sequences, producing temporally consistent and more accurate motion information for temporal alignment.

4.4. Content-Aware Modulation Fusion

Due to object occlusion and illumination changes, alignment can be unreliable in some regions. To effectively emphasize temporally consistent signals while suppressing misaligned or noisy information, we propose a content-aware modulation fusion (CMF) module (as shown in Fig. 4(b)), which adaptively fuses neighboring aligned features based on content similarity to enhance the robustness of current representation.

Specifically, CMF introduces a multi-dilation modulation (MDM) submodule that evaluates the similarity between each aligned feature and current feature across multiple spatial receptive fields, and then assigns spatially adaptive modulation parameters to the aligned feature, as illustrated in Fig. 4(c). For each aligned feature $\tilde{F}_{i-n}$, MDM

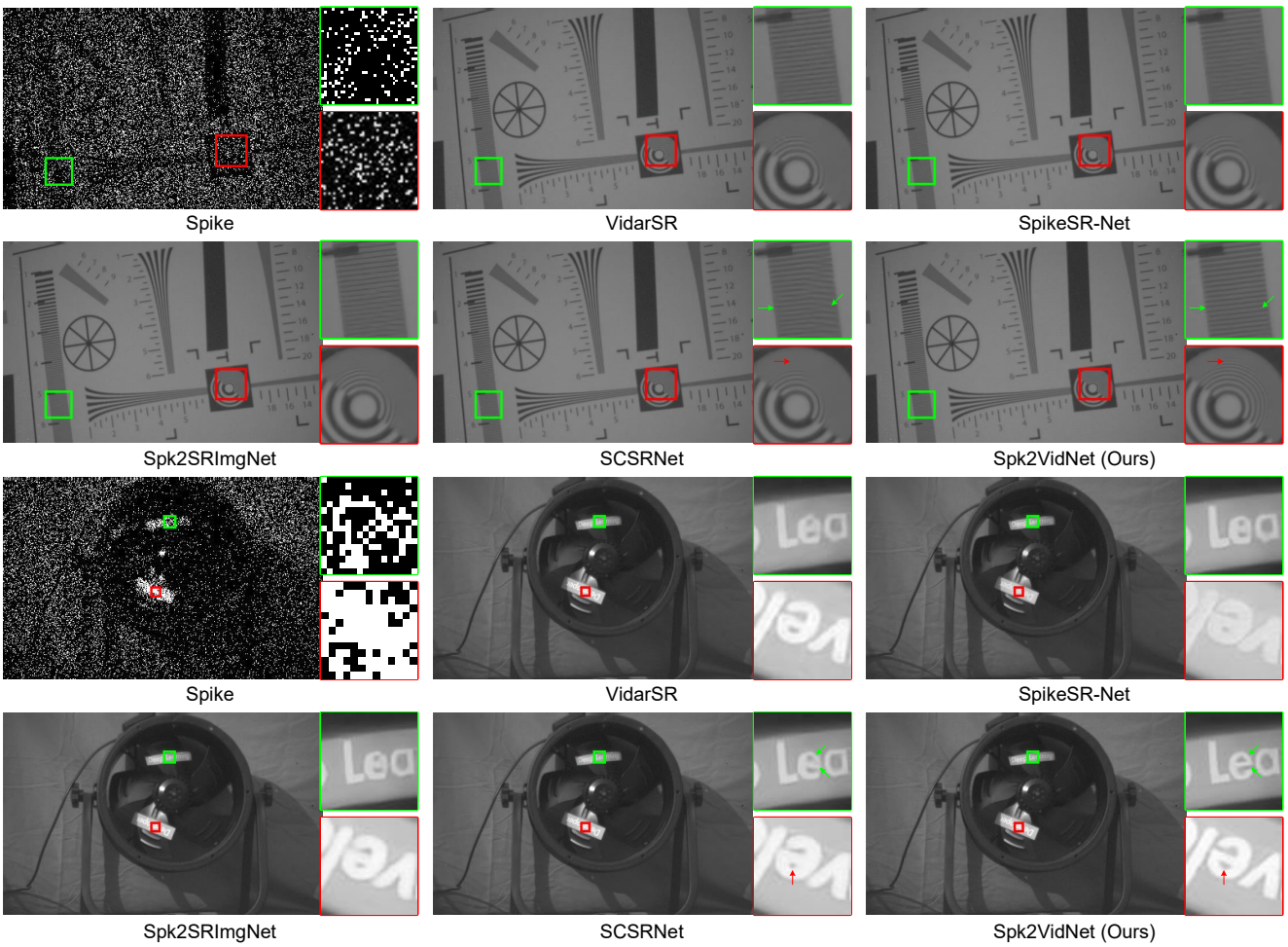


Figure 5. Visual comparison ($\times 4$) on real-captured spike data. The first scene is a resolution chart recorded by a high-speed moving spike camera. The second scene is a fast rotating fan. Please enlarge the figure for better comparison.

captures correlation strength at different spatial scales through varying dilated convolution branches $f_{\text{MDConv}}(\cdot)$:

$$\mathrm{P}_{i-n} = f_{\text{MDConv}}([\tilde{F}_{i-n}, Y_i]), \quad n = 1, \cdots K. \tag{8}$$

The multi-dilation design enables MDM to capture correlation degrees more accurately and generate appropriate scale and shift modulation parameters, denoted as α_{i-n} and β_{i-n}. MDM utilizes α_{i-n} and β_{i-n} to modulate $\tilde{F}_{i-n}$ with a residual connection:

$$\hat{F}_{i-n} = (\alpha_{i-n} \odot \tilde{F}_{i-n} + \beta_{i-n}) + \tilde{F}_{i-n}, \quad n = 1, \cdots K, \tag{9}$$

where $\odot$ denotes element-wise multiplication. Then, the modulated aligned features $\{\hat{F}_{i-n}\}_{n=1}^{K}$, together with current feature Y_i are concatenated and fed into several residual blocks $f_D(\cdot)$ to aggregate temporal correlated information:

$$F_i = f_D([\hat{F}_{i-K}, \ldots, \hat{F}_{i-1}, Y_i]), \tag{10}$$

where F_i is the updated current feature with enhanced representation, which is stored in the Layer-I feature buffer to support the update of subsequent features.

4.5. Segment-Wise Training and State Transfer

Exploiting recurrent architecture, Spk2VidNet sequentially propagates information along the temporal dimension to model long-term dependencies, which benefits super-resolution reconstruction by utilizing rich intensity information within spike streams. However, training directly on entire long spike sequences is infeasible due to limited GPU memory. To address this issue, we propose segment-wise training with state transfer strategy. Specifically, the long spike sequence is divided into multiple shorter segments, and Spk2VidNet is trained segment by segment in a sequential manner. For consecutive segments from the same sequence, the last several states of the preceding segment are detached from the computation graph, stored in the feature buffer, and then transferred as extended context to the subsequent segment, as illustrated in Fig. 6.

During training, although gradients are backpropagated only within each segment, the transferred states enable

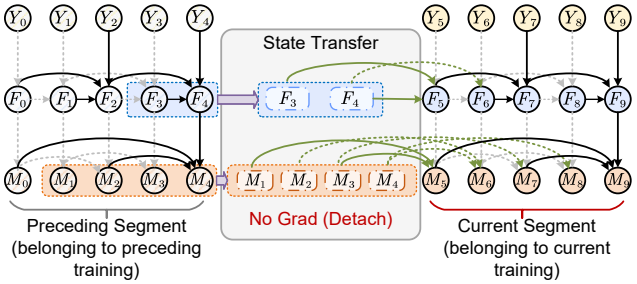


Figure 6. Segment-wise training and state transfer for modeling long temporal dependencies, illustrated with $K = 2$ as an example.

the model to exploit historical temporal information, effectively extending its modeling range. This strategy significantly reduces computational and memory costs compared to full-sequence training, while preserving temporal continuity across segments and improving reconstruction quality at segment boundaries. With this design, Spk2VidNet can efficiently handle arbitrarily long spike sequences.

5. Experiments

5.1. Datasets

Synthetic Datasets. Training SCSR models demands specialized datasets, yet capturing synchronized high-quality ground-truth images in real high-speed scenes remains difficult. Thus, current SCSR datasets [30, 31, 39, 41] are mostly simulated using the spike-camera simulator. However, these datasets usually consist of short spike sequences with limited temporal coverage, making them inadequate for training networks that aim to model long-term temporal dependencies. To address this issue, we use the spike-camera simulator to generate long spike sequence super resolution (LSSR) datasets for training and evaluation.

The LSSR datasets are built from high-frame-rate video datasets. Specifically, we employ REDS dataset [19] with 120fps and spatial resolution of 720×1280 for generating the REDS-LSSR training and evaluation dataset. REDS dataset is divided into 240 training scenes and 30 testing scenes. We follow the original partition. For each scene, we choose one long clip, yielding 240 training samples and 30 evaluation samples. Each sample in the REDS-LSSR dataset consists of one LR spike stream of size $180\times320\times461(\times4)$ / $90\times160\times461(\times8)$ (461 is the temporal length) and 45 ground truth images of size 720×1280, where the ground truth images correspond to 45 moments within LR spike sequence.

To evaluate the generalization capability of the proposed model, we generate another evaluation dataset with longer spike sequences from the Adobe240 dataset [25]. Adobe240 consists of 133 handheld-camera videos captured at 240 fps with spatial resolution of 720×1280. We adopt the test set split used in [34], which contains 17

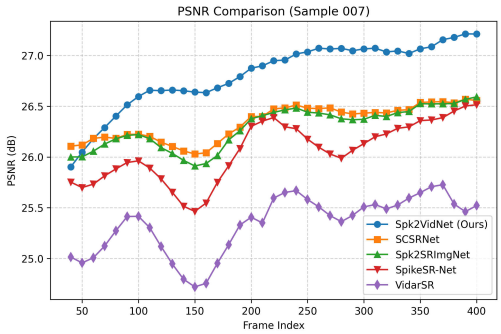


Figure 7. PSNR comparison between existing SCSR methods and the proposed Spk2VidNet on a sample from the REDS-LSSR evaluation dataset.

scenes. For each scene, we extract one long clip, resulting in 17 evaluation samples. Each sample in the Adobe240-LSSR dataset consists of one LR spike stream of size $180 \times 320 \times 911(\times4)$ / $90 \times 160 \times 911(\times8)$ and 90 corresponding HR images of size 720×1280. The sequence length of each sample in Adobe240-LSSR is approximately twice that of the sequences in REDS-LSSR evaluation set.

Real-World Captured Data. Besides the synthetic data, we also evaluate our model on real-captured spike data from publicly available datasets [44, 46] (with 400×250 spatial resolution, containing high-speed motion of objects or cameras). As ground truth references are unavailable, the evaluation is conducted through qualitative comparison.

5.2. Implementation Details

In implementation, we choose $K = 2$, meaning that during feature propagation, the current feature is aggregated with the features from two preceding moments. During training, the temporal length of input spike stream is 461, i.e., $L = 461$. The number of divided overlapping short-term spike blocks is 45, i.e., $N = 45$. Due to the long sequence length, training directly on the full sequence would exceed the available GPU memory. To address this, we partition input sequence into 5 segments, each with a length of 101 and containing 9 short-term spike blocks ($w = 10$). In first-layer propagation, the last two states from the preceding segment are transferred as additional inputs to the next segment. In second-layer propagation, the last four states from the preceding segment are passed to the subsequent segment. Our model is trained using the Adam optimizer [14] with $\mathcal{L}_1$ loss for 800 epochs. The batch size is set to 8. The initial learning rate is 0.0002 and decays by a factor of 0.7 every 100 epochs. The spike input is randomly cropped to a spatial size of 64×64. Spikes and ground truth images are randomly flipped or rotated for data augmentation. All models are trained and tested on NVIDIA RTX 3090 GPU.

5.3. Comparison with Existing Methods

We compare the proposed Spk2VidNet with existing SCSR methods: VidarSR [33], SpikeSR-Net [41], Spk2SRImgNet

Table 1. Quantitative results on REDS-LSSR and Adobe240-LSSR evaluation dataset. Red and blue indicate the best and the second-best performance, respectively. The runtime and GPU memory cost are tested on input spike stream of sizes $180 \times 320 \times 461$ ($\times 4$) and $90 \times 160 \times 461$ ($\times 8$), respectively.

Scale	Method	REDS-LSSR			Adobe240-LSSR			Params(M)	Runtime(ms)	Memory(M)
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓			
× 4	VidarSR	28.42	0.7973	0.3244	30.07	0.8806	0.2168	12.79	740	6462
	SpikeSR-Net	29.20	0.8249	0.2962	31.14	0.9040	0.1959	3.34	1088	5724
	Spk2SRImgNet	29.46	0.8346	0.2813	31.31	0.9084	0.1869	3.86	219	8248
	SCSRNet	29.50	0.8358	0.2786	31.31	0.9080	0.1867	5.30	187	4790
	Spk2VidNet (Ours)	29.92	0.8481	0.2624	32.36	0.9264	0.1607	3.73	43	4606
× 8	VidarSR	25.74	0.6851	0.4262	26.28	0.7477	0.3305	20.33	684	6142
	SpikeSR-Net	25.68	0.6796	0.4402	26.13	0.7387	0.3508	3.59	923	5550
	Spk2SRImgNet	25.80	0.6861	0.4312	26.20	0.7434	0.3442	4.01	76	4416
	SCSRNet	25.81	0.6865	0.4311	26.15	0.7426	0.3433	5.45	61	4450
	Spk2VidNet (Ours)	26.20	0.7024	0.4149	27.19	0.7847	0.3077	3.88	21	4488

[31], and SCSRNet [30], all of which are limited to handling fixed-length spike sequences. For a fair comparison, we consider only the central portion of the ground-truth image sequences of each test sample, as the compared methods are unable to reconstruct the initial and final boundary frames. We conduct experiments on $\times 4$ and $\times 8$ SR scales. We apply PSNR, SSIM [32], and LPIPS [37] to quantitatively evaluate the reconstruction quality. As shown in Tab. 1, on both evaluation datasets, our method consistently achieves the best performance while offering the fastest inference speed. To provide a more intuitive comparison, we present PSNR curves of a sample from the REDS-LSSR evaluation dataset, as shown in Fig. 7. As the frame index increases, Spk2VidNet is able to exploit long-range temporal information, leading to higher PSNR values. Besides, we conduct experiments on real-world captured spike data. As shown in Fig. 5, our method can restore clearer textures and more realistic details. More results and analysis are provided in the supplementary material.

6. Ablation Study

To verify the effectiveness of the module design in Spk2VidNet, we perform ablation study with REDS-LSSR evaluation dataset on $\times 4$ SR scale, as presented in Tab. 2. Here, we evaluate the entire HR reconstruction sequences for each sample. Case b-5 corresponds to the final model. In case b-1, we remove the MFCA and CMF modules from Spk2VidNet to form the baseline. Case b-2 adds the independent alignment to the baseline. Case 3 adds MFCA module to the baseline, while case b-4 adds CMF module to the baseline. For each b- case, the corresponding a- case represents the one-layer propagation version of that model. First, by comparing cases a-1 to a-5 and b-1 to b-5, we observe that both MFCA and CMF improve SR performance, and their combination achieves the best performance. Second, increasing the number of propagation layers from 1 (cases a-) to 2 (cases b-) further enhances the reconstruction performance.

Table 2. Ablation Study of the proposed Spk2VidNet on REDS-LSSR evaluation dataset ($\times 4$).

Case	MFCA	CMF	Propagation Layer	PSNR ↑	SSIM ↑
a-1			1	28.37	0.7950
a-2	◊		1	29.02	0.8177
a-3	✓		1	29.31	0.8300
a-4		✓	1	29.23	0.8284
a-5	✓	✓	1	29.66	0.8383
b-1			2	28.39	0.7969
b-2	◊		2	29.02	0.8197
b-3	✓		2	29.41	0.8325
b-4		✓	2	29.27	0.8273
b-5	✓	✓	2	**29.79**	**0.8432**

We also evaluate the impact of segment-wise training with state transfer. When each segment is trained independently, without transferring states across segments, the SR reconstruction achieves 29.58 dB / 0.8377 PSNR/SSIM, which is lower than the performance achieved using the proposed training strategy (29.79 dB/ 0.8432). More ablation study is provided in the supplementary material.

7. Conclusion

This paper develops a hierarchical recurrent network named Spk2VidNet to reconstruct HR image sequences from long spike streams. To handle spike fluctuations, we design an architecture that performs two-layer propagation with enlarged temporal receptive fields, enabling progressive refinement of feature representations. Within the propagation block, we introduce a multi-frame consistency guided alignment (MFCA) module and a content-aware modulation fusion (CMF) module to adaptively aggregate temporal information, enhancing temporally consistent signals and suppressing disturbances. To model long-term dependencies with limited GPU memory, we propose a segment-wise training with state transfer strategy. Experimental results demonstrate that the proposed method achieves state-of-the-art performance on synthetic and real-captured spike data.

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